

java.io.RandomAccessFile

```
RandomAccessFile(String name, String mode)
    costruttore
void seek(long pos) throws IOException
long length() throws IOException
long getFilePointer() throws IOException
void writeChars(String s) throws IOException
void writeChar(char v) throws IOException
void writeDouble(double v) throws IOException
void writeInt(int v) throws IOException
char readChar() throws IOException
double readDouble() throws IOException
int readInt() throws IOException
void close() throws IOException
```

java.io.ObjectInputStream

```
ObjectInputStream(FileInputStream in) throws IOException
Object readObject() throws IOException, ClassNotFoundException
void close() throws IOException
```

java.io.ObjectOutputStream

```
ObjectOutputStream(FileOutputStream out) throws IOException
void writeObject(Object obj) throws IOException
void close() throws IOException
```

java.io.FileInputStream

```
FileInputStream(String name) throws FileNotFoundException
```

java.io.FileOutputStream

```
FileOutputStream(String name, boolean append)
```

java.util.Hashtable<K,V>

```
Hashtable()  
boolean containsKey(Object key)  
V get(Object key)  
V put(K key, V value)  
Set<K> keySet()  
Enumeration<V> elements()
```

java.util.Iterator<E>

```
boolean hasNext()  
E next()
```

java.util.Enumeration<E>

```
Iterator<E> asIterator()
```

StringBuilder

```
setLenght(int len)
```

```
import java.io.IOException;  
import java.io.RandomAccessFile;  
import java.util.InputMismatchException;  
import java.util.Scanner;
```

```
public class Input {
```

```

public static int lgInt(int vmin, int vmax, String mex){
    Scanner in = new Scanner(System.in);
    int x = 0;
    boolean val = false;
    while(!val){
        try {
            System.out.print(mex);
            x = in.nextInt();
            while (x < vmin || x > vmax) {
                System.out.println("Errore");
                System.out.print(mex);
                x = in.nextInt();
            }
            val = true;
        } catch (InputMismatchException exc) {
            System.out.println(exc.getMessage());
            in.nextLine();
        }
    }
    return x;
}

```

```

public static double lgDbl(double vmin, double vmax, String mex){
    Scanner in = new Scanner(System.in);
    double x = 0;
    boolean val = false;
    while(!val){
        try {
            System.out.print(mex);
            x = in.nextDouble();
            while (x < vmin || x > vmax) {
                System.out.println("Errore");
                System.out.print(mex);
                x = in.nextDouble();
            }
            val = true;
        } catch (InputMismatchException exc) {
            System.out.println(exc.getMessage());
            in.nextLine();
        }
    }
}

```

```

        return x;
    }
    public static char lgChar(String mex){
        Scanner in = new Scanner(System.in);
        char c;
        System.out.print(mex);
        c = in.next().charAt(0);
        while(c == '\0'){
            System.out.println("errore");
            System.out.print(mex);
            c = in.next().charAt(0);
        }
        return c;
    }
    public static String lgStr(String mex){
        Scanner in = new Scanner(System.in);
        String s;
        System.out.print(mex);
        s = in.next();
        while(s.isEmpty()){
            System.out.println("Errore");
            System.out.print(mex);
            s = in.next();
        }
        return s;
    }
    public static void writeString(RandomAccessFile raf, String str, int len) throws
IOException {
        StringBuilder buf = new StringBuilder(str);
        buf.setLength(len);
        raf.writeChars(buf.toString());
    }
    public static String readString(RandomAccessFile raf, int len) throws IOException{
        char[] str = new char[len];
        for(int k = 0; k < len; k++)
            str[k] = raf.readChar();

        String s = new String(str);
        return s.trim();
    }
}

```